

Measuring ECG through Defibrillation Electrodes: A Comprehensive Analysis of Signal Acquisition, Processing, and Noise Mitigation Strategies

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Abstract

Continuous electrocardiogram (ECG) monitoring during cardiac resuscitation is essential for assessing a patient’s rhythm before and after defibrillation therapy. However, the high-energy electrical pulses delivered during defibrillation—typically reaching 5 kV and 50 A—introduce severe artifacts that saturate conventional ECG amplifiers and obscure the underlying cardiac signal. This report presents a comprehensive analysis of the challenges, circuit protection strategies, and advanced signal processing methodologies required to measure diagnostic-quality ECG signals through the same electrodes used for defibrillation. The system architecture incorporates a multi-stage hardware protection front-end compliant with IEC 60601-2-4, a high-CMRR instrumentation amplifier with Right Leg Drive, and a digital signal processing pipeline that integrates adaptive filtering algorithms (LMS, NLMS, RLS) with discrete wavelet transform denoising. Additionally, the feasibility of a multi-functional defibrillation electrode integrating ECG, temperature, and blood oxygen saturation sensors is evaluated. Analysis demonstrates that electrode polarization recovery within 400 mV at 4 s post-shock is achievable with Ag/AgCl electrodes conforming to ANSI/AAMI EC12 standards. The report concludes with a discussion of emerging machine learning approaches, including CNN-LSTM architectures, for real-time cardiac rhythm classification during resuscitation.

1 Introduction

Electrocardiogram (ECG) monitoring is a fundamental diagnostic tool in modern cardiac care, providing real-time visualization of the heart’s electrical activity through the characteristic P-QRS-T waveform complex. During cardiac arrest and resuscitation scenarios, continuous ECG monitoring becomes critically important—clinicians must rapidly assess the patient’s cardiac rhythm to determine whether defibrillation is indicated and to evaluate the effectiveness of delivered shocks [1].

In current clinical practice, defibrillation and ECG monitoring employ separate electrode systems. Standard 12-lead ECG systems use ten small-area adhesive electrodes placed at specific anatomical positions, while defibrillation requires large-area electrode pads positioned in anterior-lateral (AL) or anterior-posterior (AP) configurations [2]. This dual-electrode approach introduces several clinical complications: increased setup time during time-critical emergencies, greater equipment complexity, and potential interference between electrode systems.

The concept of using defibrillation electrodes simultaneously for ECG acquisition offers significant clinical advantages—simplified patient preparation, reduced cable clutter,

and uninterrupted monitoring during shock delivery. However, this approach presents formidable engineering challenges. The defibrillation pulse, which delivers energy levels of 150 J to 360 J through biphasic waveforms, creates transient voltages and currents that exceed the input range of standard ECG amplifiers by several orders of magnitude [3]. The resulting electrode polarization, amplifier saturation, and baseline drift can render the ECG signal unreadable for periods ranging from seconds to minutes after shock delivery.

This report provides a comprehensive technical analysis of the challenges and solutions associated with ECG measurement through defibrillation electrodes. The specific objectives are:

- To characterize the electrochemical mechanisms of electrode polarization following high-energy defibrillation pulses.
- To design a hardware protection and signal conditioning front-end compliant with IEC 60601-2-4 and ANSI/AAMI EC12/EC13 standards.
- To evaluate advanced digital signal processing techniques—including adaptive filtering, wavelet denoising, and machine learning—for artifact removal and signal recovery.
- To assess the feasibility of multifunctional defibrillation electrodes integrating multiple physiological sensors.

2 Background and Literature Review

2.1 ECG Signal Characteristics

The electrocardiogram represents the summation of cardiac action potentials as recorded at the body surface. The standard ECG signal has the following characteristics relevant to this work [4]:

Table 1: ECG Signal Characteristics

Parameter	Typical Value	Relevance
Amplitude (QRS)	0.5 mV–5 mV	Dynamic range requirement
Frequency bandwidth	0.05 Hz–150 Hz	Filter design constraints
P-wave amplitude	0.1 mV–0.3 mV	Sensitivity requirement
T-wave amplitude	0.1 mV–0.5 mV	Low-frequency preservation
Heart rate range	30–300 bpm	Temporal resolution
DC offset (skin-electrode)	± 300 mV	Baseline management

The diagnostic bandwidth of 0.05 Hz to 150 Hz is particularly important, as the low-frequency components overlap significantly with defibrillation-induced baseline drift artifacts. The monitoring bandwidth, used in emergency settings, is typically 0.5 Hz to 40 Hz, which provides adequate rhythm assessment while improving noise rejection [5].

2.2 Defibrillation Principles

Modern external defibrillators deliver biphasic truncated exponential (BTE) or rectilinear biphasic (RLB) waveforms. The key electrical parameters are:

- **Peak voltage:** 1500 V–5000 V (depending on impedance compensation)
- **Peak current:** up to 50 A
- **Delivered energy:** 150 J–360 J (biphasic)
- **Pulse duration:** 4 ms–20 ms per phase
- **Patient impedance:** 25 Ω –180 Ω (typical range)

The instantaneous power delivered during a defibrillation shock can exceed 250 kW, creating an electromagnetic environment that poses extreme challenges for simultaneous low-level biopotential measurement [6].

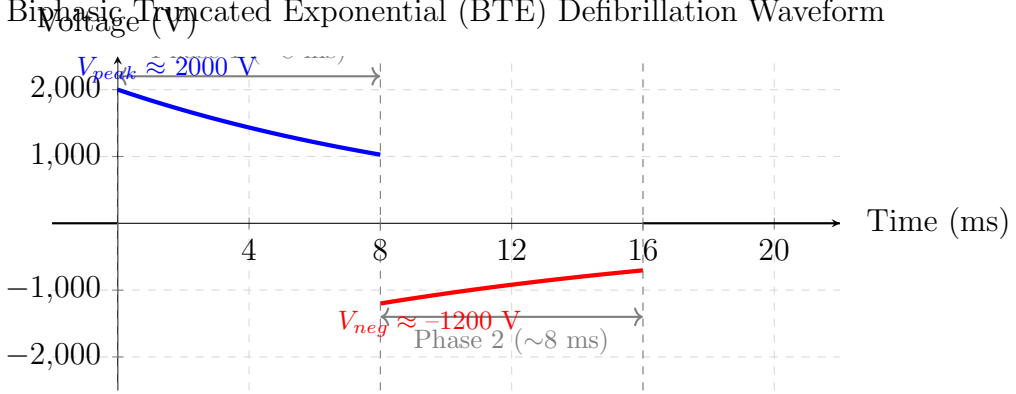


Figure 1: Biphasic truncated exponential (BTE) defibrillation waveform showing the positive Phase 1 and negative Phase 2, each approximately 8 ms in duration. Peak voltage depends on patient impedance and energy setting.

2.3 Electrode Polarization Physics

When defibrillation current flows through an ECG electrode, the electrode-electrolyte interface undergoes forced electrochemical reactions that shift the half-cell potential away from its equilibrium value. This phenomenon, termed electrode polarization, is governed by the Butler-Volmer equation [4]:

$$i = i_0 \left[\exp \left(\frac{\alpha_a n F \eta}{RT} \right) - \exp \left(-\frac{\alpha_c n F \eta}{RT} \right) \right] \quad (1)$$

where i_0 is the exchange current density, α_a and α_c are the anodic and cathodic transfer coefficients, n is the number of electrons transferred, F is Faraday’s constant, η is the overpotential, R is the gas constant, and T is the absolute temperature.

2.3.1 Ag/AgCl Electrode Recovery

Silver/silver chloride (Ag/AgCl) electrodes are the clinical standard for biopotential measurement due to their near-ideal non-polarizable behavior. The reversible redox reaction at the electrode surface is:



This reaction provides fast electrode kinetics with low exchange impedance, enabling rapid recovery from defibrillation-induced polarization. However, extreme or repeated high-intensity currents can degrade the AgCl coating, progressively increasing polarization potential and impairing signal quality [7].

2.3.2 Polarization Recovery Standards

The recovery of electrode potential following defibrillation is quantified by the ANSI/AAMI EC12:2000/(standard. Figure 2 illustrates the characteristic recovery curves for Ag/AgCl and stainless steel electrodes, with the AAMI compliance limits overlaid.

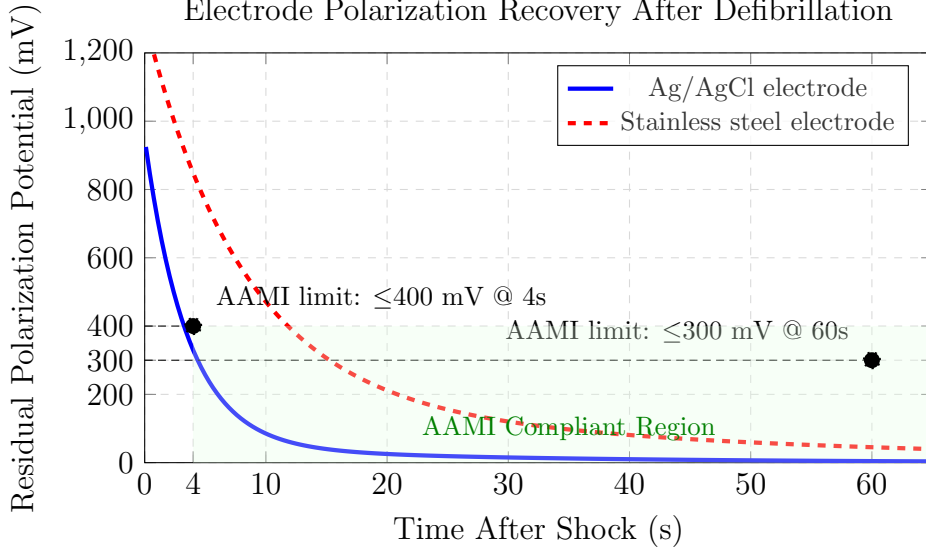


Figure 2: Polarization potential recovery curves for Ag/AgCl and stainless steel electrodes following a defibrillation shock. Ag/AgCl electrodes recover within ANSI/AAMI EC12 limits (400 mV at 4 s, 300 mV at 60 s), while stainless steel electrodes exhibit prolonged recovery.

The standard specifies the following performance limits for disposable ECG electrodes [8]:

Table 2: ANSI/AAMI EC12 Polarization Recovery Requirements

Time After Final Shock	Maximum Residual Potential
4 s	≤ 400 mV
60 s	≤ 300 mV

The clinical objective is to restore visible ECG signal within 5 s to 10 s after shock delivery, enabling clinicians to assess the post-shock rhythm immediately [9].

2.4 Regulatory Framework

The design of ECG monitoring systems integrated with defibrillators must comply with multiple international standards:

- **IEC 60601-1:** General requirements for basic safety and essential performance of medical electrical equipment.
- **IEC 60601-2-4:2010+AMD1:2018:** Particular requirements for cardiac defibrillators, including defibrillation-proof applied parts [10].

- **ANSI/AAMI EC12:2000/(R)2020:** Disposable ECG electrode performance requirements, including defibrillation overload recovery testing [8].
- **ANSI/AAMI EC13:2002/(R)2020:** Cardiac monitor, heart rate meter, and alarm requirements, specifying defibrillator overload protection for monitoring devices [9].

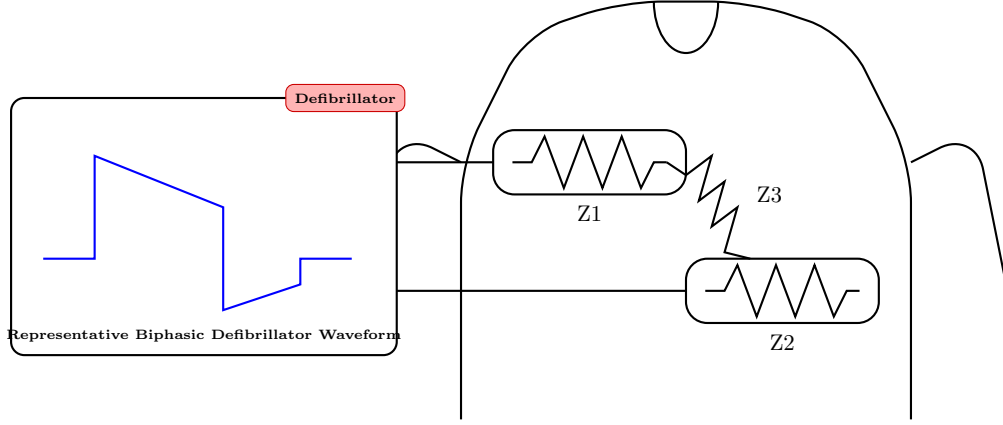


Figure 3: Defibrillation electrode impedance model showing the electrode-skin contact impedances ($Z1$, $Z2$) and the transthoracic impedance ($Z3$) connected to a biphasic defibrillator.

3 System Architecture

The proposed system for ECG measurement through defibrillation electrodes comprises three primary subsystems: (1) hardware protection and input conditioning, (2) analog signal amplification and filtering, and (3) digital signal processing. Each subsystem is designed to address specific challenges arising from the coexistence of high-energy defibrillation pulses and microvolt-level cardiac signals. The complete signal flow is illustrated in Figure 4.

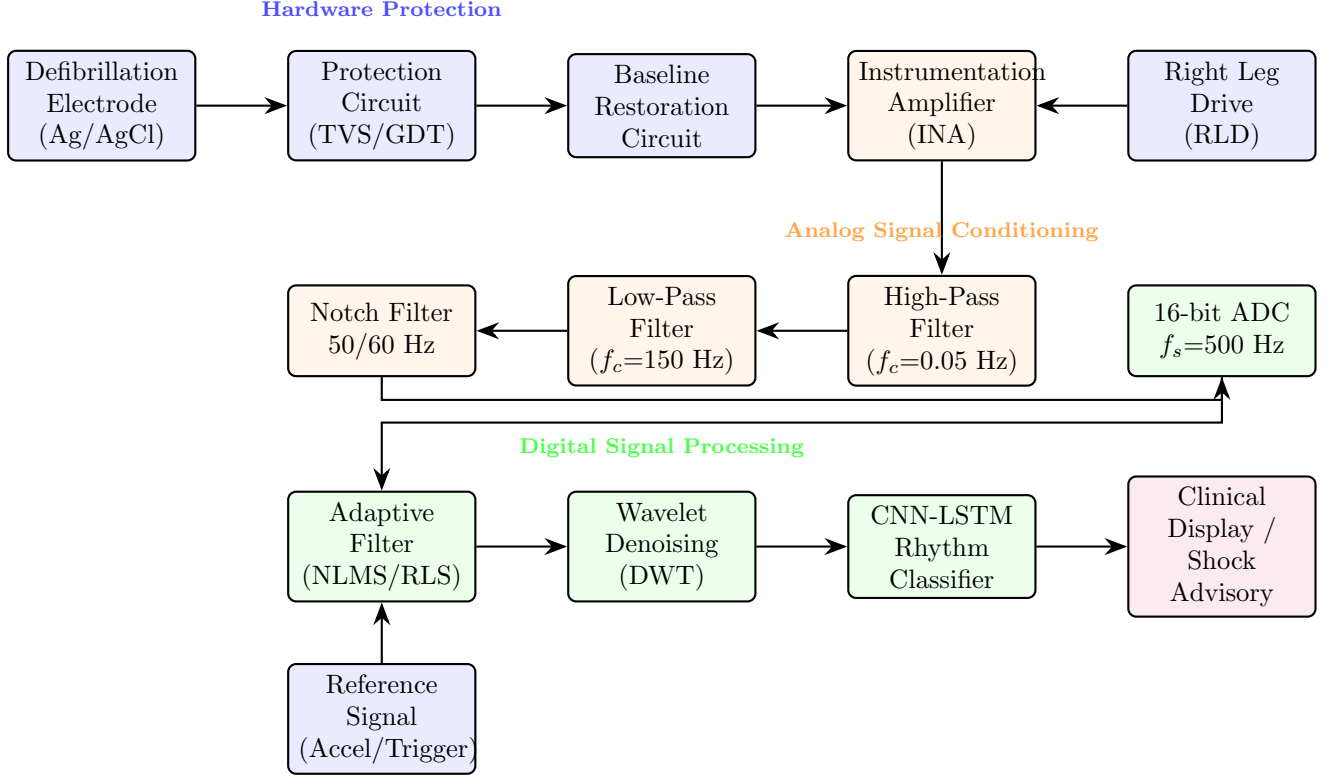


Figure 4: Complete system block diagram for ECG measurement through defibrillation electrodes, showing the signal flow from electrode acquisition through hardware protection, analog conditioning, and digital signal processing to clinical output.

3.1 Hardware Protection Circuit

The protection circuit serves as the first line of defense against defibrillation pulse damage. Under IEC 60601-2-4, the ECG monitoring inputs must be “defibrillation-proof,” meaning they must withstand the full energy of the defibrillation pulse without sustaining permanent damage and must recover normal monitoring function within the time limits specified by ANSI/AAMI EC13 [10].

The protection circuit employs a multi-stage approach:

1. **Current-Limiting Resistors:** High-power wirewound resistors (1 k Ω –10 k Ω , rated for pulse energy dissipation) are placed in series with each input lead. For a 5 kV pulse across 10 k Ω series resistance, the peak current is limited to 500 mA, protecting downstream components.
2. **Voltage Clamping:** Bidirectional transient voltage suppressor (TVS) diodes or gas discharge tubes (GDTs) clamp the voltage to levels safe for the instrumentation amplifier. GDTs offer higher surge current handling (>10 kA) but slower response (1 μ s), while TVS diodes provide nanosecond-scale clamping [11].
3. **Baseline Restoration:** Following a defibrillation pulse, the high-pass filter capacitors in the signal path accumulate charge, causing prolonged baseline saturation. Electronically switchable filter time constants—implemented via analog switches or MOSFET-based reset circuits—rapidly discharge this stored energy, reducing recovery time from minutes to milliseconds [12].

The circuit-level implementation of this protection strategy is illustrated in Figure 5.

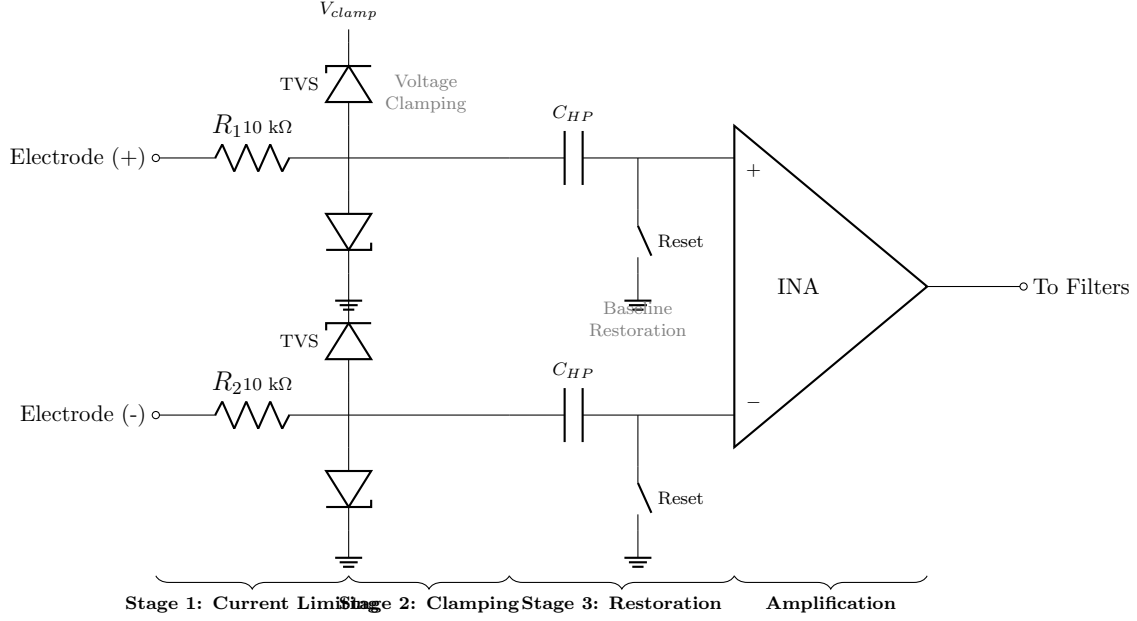


Figure 5: Hardware protection circuit schematic showing the three-stage protection approach: (1) current-limiting series resistors ($R_1, R_2 = 10 \text{ k}\Omega$), (2) bidirectional TVS diode voltage clamping, and (3) switchable baseline restoration with MOSFET-controlled capacitor discharge, followed by the instrumentation amplifier (INA).

3.2 Analog Front-End Design

3.2.1 Instrumentation Amplifier

The core of the analog front-end is a high-performance instrumentation amplifier (INA) configured for differential ECG signal acquisition. Critical specifications include:

Table 3: Instrumentation Amplifier Specifications

Parameter	Requirement
Common-Mode Rejection Ratio (CMRR)	$\geq 110 \text{ dB}$ at 50 Hz/60 Hz
Input impedance	$\geq 10 \text{ M}\Omega$
Input bias current	$\leq 10 \text{ nA}$
Gain	100–1000 (programmable)
Input-referred noise	$\leq 1 \mu\text{V}_{\text{pp}}$ (0.05 Hz–150 Hz)
Defibrillation overload recovery	$< 1 \text{ s}$ to $\pm 10 \%$ of baseline

The differential voltage V_{ECG} measured between two defibrillation electrodes is:

$$V_{\text{ECG}} = A_d(V_+ - V_-) + A_{cm} \left(\frac{V_+ + V_-}{2} \right) \quad (3)$$

where A_d is the differential gain and A_{cm} is the common-mode gain. The CMRR is defined as:

$$\text{CMRR} = 20 \log_{10} \left(\frac{A_d}{A_{cm}} \right) [\text{dB}] \quad (4)$$

For power-line interference at 50 Hz with common-mode amplitude of 1 V, a CMRR of 110 dB reduces the interference to approximately 3.2 μV —below the noise floor of the system.

3.2.2 Right Leg Drive (RLD) Circuit

The Right Leg Drive circuit actively suppresses common-mode interference by sensing the average common-mode voltage at the patient and driving an inverted, amplified version back through a reference electrode. The transfer function of the RLD circuit is [13]:

$$V_{\text{RLD}} = -\frac{R_f}{R_a} \cdot V_{cm} \quad (5)$$

where R_f is the feedback resistance, R_a is the averaging resistance, and V_{cm} is the common-mode voltage. Typical implementations achieve an additional 20 dB–40 dB of common-mode rejection beyond the instrumentation amplifier’s inherent CMRR.

3.2.3 Analog Filter Stages

The analog filter chain consists of three cascaded stages:

1. **High-Pass Filter (HPF):** Second-order Butterworth with $f_c = 0.05$ Hz (diagnostic mode) or $f_c = 0.5$ Hz (monitoring mode). Transfer function:

$$H_{\text{HPF}}(s) = \frac{s^2}{s^2 + \frac{\omega_c}{Q}s + \omega_c^2} \quad (6)$$

where $\omega_c = 2\pi f_c$ and $Q = 0.707$ for Butterworth response.

2. **Low-Pass Filter (LPF):** Fourth-order Butterworth with $f_c = 150$ Hz (diagnostic) or $f_c = 40$ Hz (monitoring), providing -80 dB/decade roll-off to attenuate EMG and high-frequency noise.
3. **Notch Filter:** Twin-T or active notch filter centered at 50 Hz (or 60 Hz) with $Q = 10$ –30, providing ≥ 30 dB rejection of power-line interference. The transfer function is:

$$H_{\text{notch}}(s) = \frac{s^2 + \omega_n^2}{s^2 + \frac{\omega_n}{Q}s + \omega_n^2} \quad (7)$$

The combined frequency response of the analog filter chain is shown in Figure 6.

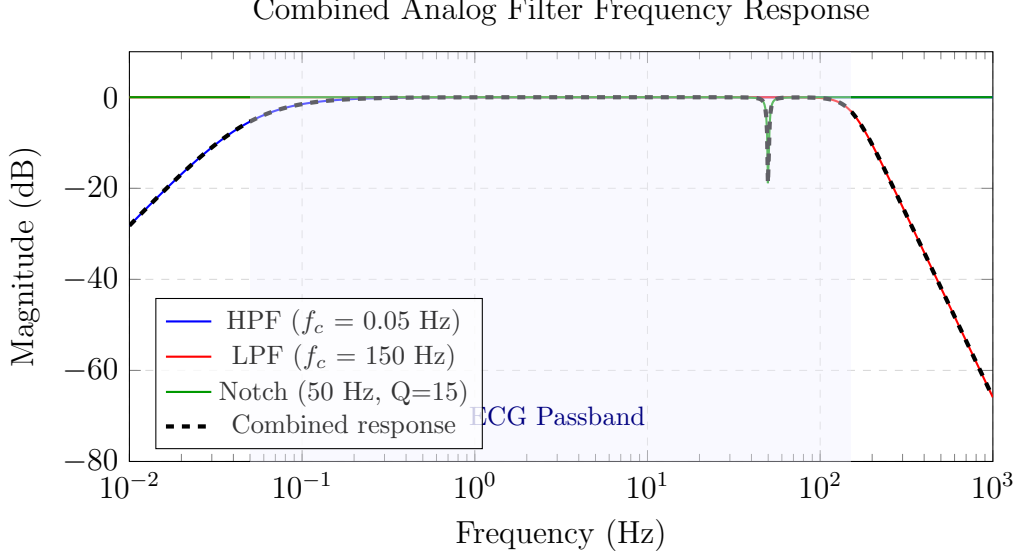


Figure 6: Bode magnitude plot of the analog filter chain showing the high-pass filter (0.05 Hz), low-pass filter (150 Hz), and 50 Hz notch filter responses. The dashed line shows the combined response, with the ECG diagnostic passband highlighted.

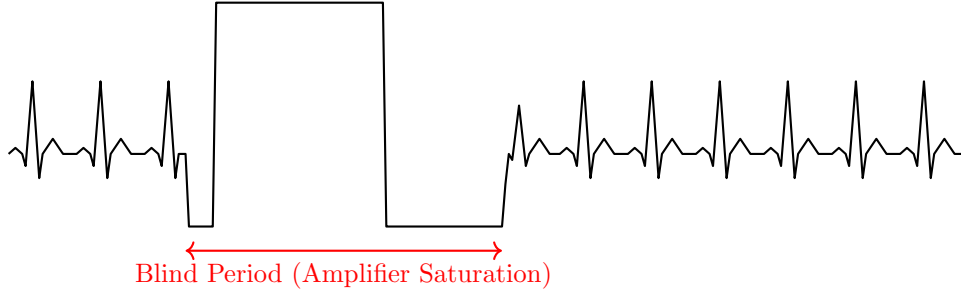


Figure 7: ECG waveform distortion during and after defibrillation shock delivery, illustrating the “blind period” where cardiac rhythm assessment is impossible due to amplifier saturation and electrode polarization.

4 Signal Processing Methodology

Following analog-to-digital conversion (typically 16-bit at 500 Hz–1000 Hz sampling rate), digital signal processing techniques are employed to further enhance signal quality and extract clinically relevant information from the corrupted ECG signal.

4.1 Adaptive Filtering

Adaptive filters are particularly effective for removing defibrillation artifacts and CPR-related interference because they can dynamically adjust their coefficients to track non-stationary noise characteristics [14].

4.1.1 Least Mean Squares (LMS) Algorithm

The LMS algorithm minimizes the mean square error between the desired signal and the filter output. The weight update equation is:

$$\mathbf{w}(n+1) = \mathbf{w}(n) + 2\mu e(n) \mathbf{x}(n) \quad (8)$$

where $\mathbf{w}(n)$ is the filter weight vector, μ is the step size (learning rate), $e(n) = d(n) - y(n)$ is the error signal, and $\mathbf{x}(n)$ is the input reference vector. The step size must satisfy the stability condition:

$$0 < \mu < \frac{1}{\lambda_{\max}} \quad (9)$$

where $\lambda_{\max}$ is the largest eigenvalue of the input autocorrelation matrix $\mathbf{R}_{xx}$. For ECG denoising applications, typical step sizes range from 10^{-4} to 10^{-2} [14].

4.1.2 Normalized LMS (NLMS) Algorithm

The NLMS algorithm provides improved convergence by normalizing the step size with respect to the input signal power:

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \frac{\mu}{\|\mathbf{x}(n)\|^2 + \epsilon} e(n) \mathbf{x}(n) \quad (10)$$

where ϵ is a small positive constant for numerical stability. The NLMS algorithm converges faster than standard LMS for non-stationary signals, making it well-suited for defibrillation artifact removal where the noise statistics change rapidly [15].

4.1.3 Recursive Least Squares (RLS) Algorithm

The RLS algorithm offers superior convergence speed and tracking capability at the cost of increased computational complexity ($O(N^2)$ per sample versus $O(N)$ for LMS). The weight update is [14]:

$$\mathbf{w}(n) = \mathbf{w}(n-1) + \mathbf{k}(n) e(n) \quad (11)$$

where the gain vector $\mathbf{k}(n)$ is computed as:

$$\mathbf{k}(n) = \frac{\mathbf{P}(n-1)\mathbf{x}(n)}{\lambda + \mathbf{x}^T(n)\mathbf{P}(n-1)\mathbf{x}(n)} \quad (12)$$

and $\mathbf{P}(n)$ is the inverse correlation matrix updated recursively with forgetting factor λ (typically $0.95 \leq \lambda \leq 1.0$).

Table 4: Comparison of Adaptive Filtering Algorithms for ECG Denoising

Property	LMS	NLMS	RLS
Computational complexity	$O(N)$	$O(N)$	$O(N^2)$
Convergence speed	Slow	Moderate	Fast
Tracking capability	Limited	Good	Excellent
Numerical stability	High	High	Moderate
Real-time suitability	Excellent	Excellent	Good
Typical SNR improvement	8 dB–12 dB	10 dB–15 dB	15 dB–20 dB

In practical implementations, a reference signal correlated with the noise but uncorrelated with the ECG is required. For CPR artifact removal, chest compression accelerometer signals serve as effective references. For defibrillation artifact removal, the defibrillation trigger signal and electrode impedance measurements provide suitable reference inputs [16].

The adaptive noise cancellation architecture is illustrated in Figure 8.

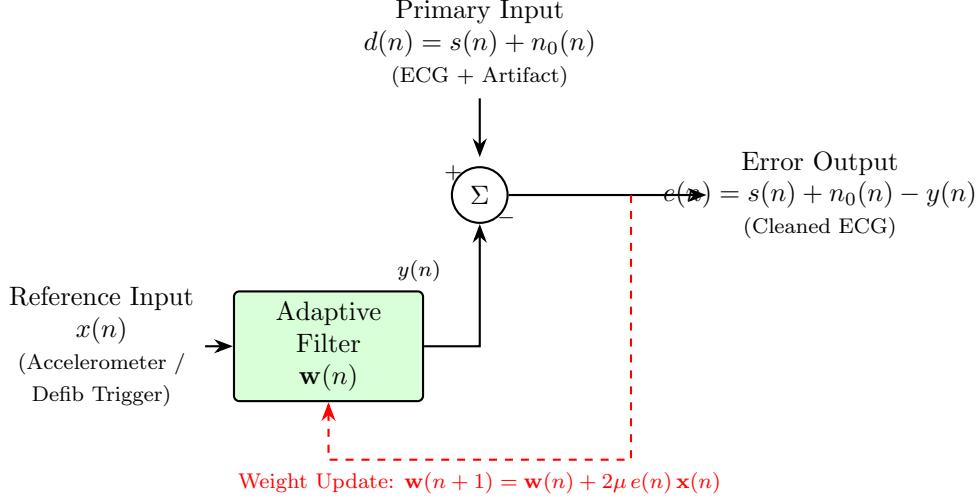


Figure 8: Adaptive noise cancellation architecture for ECG defibrillation artifact removal. The reference input (correlated with the noise source) drives the adaptive filter, whose output is subtracted from the primary input containing the corrupted ECG. The error signal serves as both the cleaned ECG output and the adaptation feedback signal.

4.2 Wavelet Transform Denoising

The Discrete Wavelet Transform (DWT) provides time-frequency localization that is particularly advantageous for ECG denoising, as the ECG signal is inherently non-stationary [17].

4.2.1 Decomposition

The DWT decomposes the signal $x(n)$ into approximation coefficients c_A and detail coefficients c_D at multiple resolution levels:

$$c_A[j, k] = \sum_n x(n) \phi_{j,k}(n), \quad c_D[j, k] = \sum_n x(n) \psi_{j,k}(n) \quad (13)$$

where $\phi_{j,k}$ is the scaling function and $\psi_{j,k}$ is the wavelet function at scale j and translation k .

4.2.2 Thresholding

Noise reduction is achieved by applying soft or hard thresholding to the detail coefficients. The soft thresholding function is:

$$\hat{c}_D[j, k] = \text{sign}(c_D[j, k]) \cdot \max(|c_D[j, k]| - \lambda, 0) \quad (14)$$

where λ is the threshold level, commonly estimated using the universal threshold:

$$\lambda = \sigma\sqrt{2 \ln N} \quad (15)$$

where σ is the noise standard deviation (estimated from the finest-scale detail coefficients using the median absolute deviation) and N is the signal length.

For ECG applications, the Daubechies-4 (db4) and Symlet-8 (sym8) mother wavelets have been shown to provide optimal performance due to their morphological similarity to the QRS complex [17]. Typical decomposition to level 8 at 500 Hz sampling rate provides effective separation of the ECG components from high-frequency noise and baseline drift.

The multi-level wavelet decomposition and reconstruction process is illustrated in Figure 9.

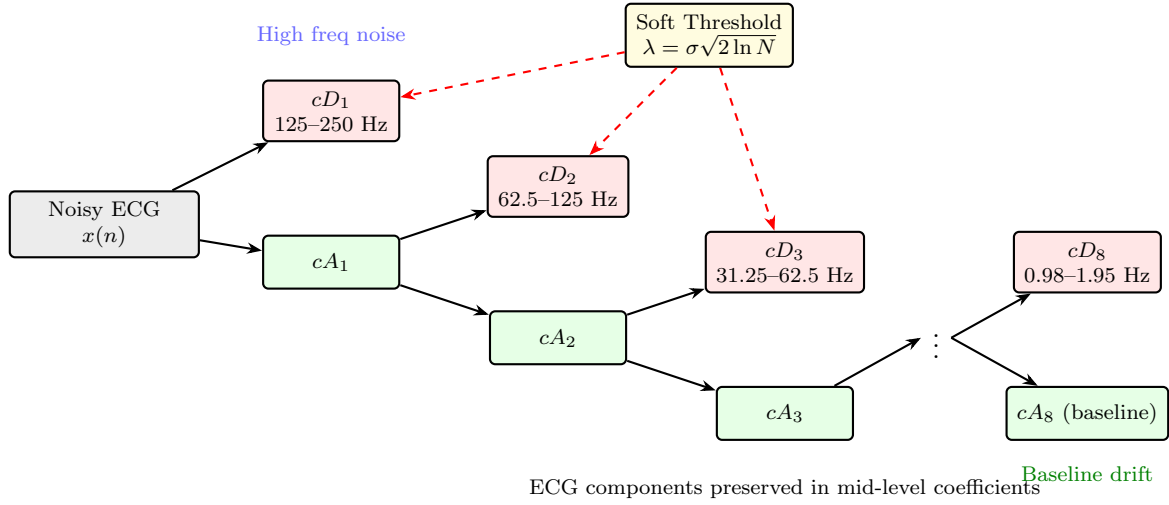


Figure 9: Multi-level discrete wavelet transform (DWT) decomposition tree for ECG denoising. The noisy signal is recursively decomposed into detail (cD) and approximation (cA) coefficients at 8 levels. Soft thresholding is applied to detail coefficients to remove noise while preserving ECG morphology. At 500 Hz sampling rate, each level halves the frequency band.

4.3 Machine Learning Approaches

Recent advances in deep learning have introduced powerful alternatives for ECG signal processing during resuscitation scenarios.

4.3.1 CNN-Based Spatial Feature Extraction

One-dimensional Convolutional Neural Networks (1D-CNNs) can automatically extract morphological features from ECG waveforms without manual feature engineering. The convolutional operation for layer l is:

$$z_k^{(l)}(n) = \sigma \left(\sum_m w_{k,m}^{(l)} * x_m^{(l-1)}(n) + b_k^{(l)} \right) \quad (16)$$

where $w_{k,m}^{(l)}$ are the learnable filter weights, $*$ denotes convolution, σ is a non-linear activation function, and $b_k^{(l)}$ is the bias term.

4.3.2 LSTM-Based Temporal Modeling

Long Short-Term Memory (LSTM) networks capture long-range temporal dependencies in cardiac rhythms that are critical for distinguishing shockable rhythms (ventricular fibrillation, pulseless ventricular tachycardia) from non-shockable rhythms. The LSTM cell state update is governed by [18]:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f) \quad (\text{forget gate}) \quad (17)$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i) \quad (\text{input gate}) \quad (18)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o) \quad (\text{output gate}) \quad (19)$$

4.3.3 Hybrid CNN-LSTM Architecture

The hybrid CNN-LSTM approach cascades CNN layers for spatial feature extraction with LSTM layers for temporal modeling, as illustrated in Figure 10. This architecture has achieved classification accuracies exceeding 98–99% on benchmark datasets such as the MIT-BIH Arrhythmia Database for rhythm classification tasks [19]. For defibrillation applications, such systems can provide automated shock advisory decisions within 100 ms–500 ms, supporting rapid clinical decision-making during resuscitation.

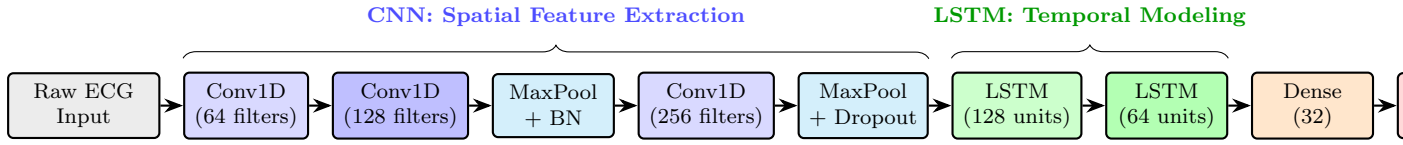


Figure 10: Hybrid CNN-LSTM architecture for automated cardiac rhythm classification. Convolutional layers extract morphological features from ECG waveforms, while LSTM layers capture temporal dependencies. The softmax output provides rhythm classification for shock advisory decisions (VF: ventricular fibrillation, VT: ventricular tachycardia, NSR: normal sinus rhythm).

5 Multifunctional Defibrillation Electrode Design

In clinical critical care settings, patients are instrumented with numerous sensors for real-time vital sign monitoring: temperature sensors, ECG electrodes, pulse oximetry probes, blood pressure transducers, and respiratory sensors. Each sensor requires its own placement site, cables, and interface connections. This configuration is cumbersome, increases nursing workload, limits patient mobility, and can contribute to iatrogenic injuries such as contact allergies and pressure ulcers [20].

A proposed solution is the multifunctional defibrillation electrode, which integrates multiple sensing modalities into the defibrillation pad assembly. The design concept comprises left and right defibrillation electrode patches, each incorporating specific sensors:

Table 5: Multifunctional Electrode Sensor Configuration

Right Defibrillation Patch	Left Defibrillation Patch
Temperature sensor (NTC thermistor)	Multiple ECG sensors (stepped array)
SpO ₂ sensor (reflective IR)	Optical fiber position sensor
Optical fiber position sensor	—
ECG sensor	—

5.1 Electrical Isolation Strategy

Each sensor is electrically isolated from the defibrillation electrode’s conductive coating through a dedicated insulating layer. In regions where sensors are positioned, the conductive coating is designed with a hollow structure, and each sensor base is covered with insulating heat-dissipating material constructed from fiber textiles. This architecture ensures that:

- The defibrillation pulse does not couple into sensor signal paths.
- Sensor measurements are not corrupted during and immediately after shock delivery.
- Thermal management prevents sensor damage from pulse energy dissipation.

5.2 Technical Challenges

Integrating optical SpO₂ measurement into a defibrillation pad presents unique challenges. Pulse oximetry requires plethysmographic (PPG) signal acquisition using specific wavelengths (red at 660 nm and infrared at 940 nm). In the reflective configuration suitable for chest-wall placement, the signal amplitude is significantly lower than finger-clip transmission-mode sensors, requiring higher gain amplification and more aggressive noise filtering [21].

6 Analysis and Discussion

6.1 Effect of Electrode Placement on ECG Morphology

The defibrillation electrode positions—anterior-lateral (AL) and anterior-posterior (AP)—do not correspond to standard diagnostic ECG lead positions. The resulting lead vectors produce signal morphologies that differ from conventional 12-lead recordings. The two standard placement configurations and their corresponding lead vectors are illustrated in Figure 11.

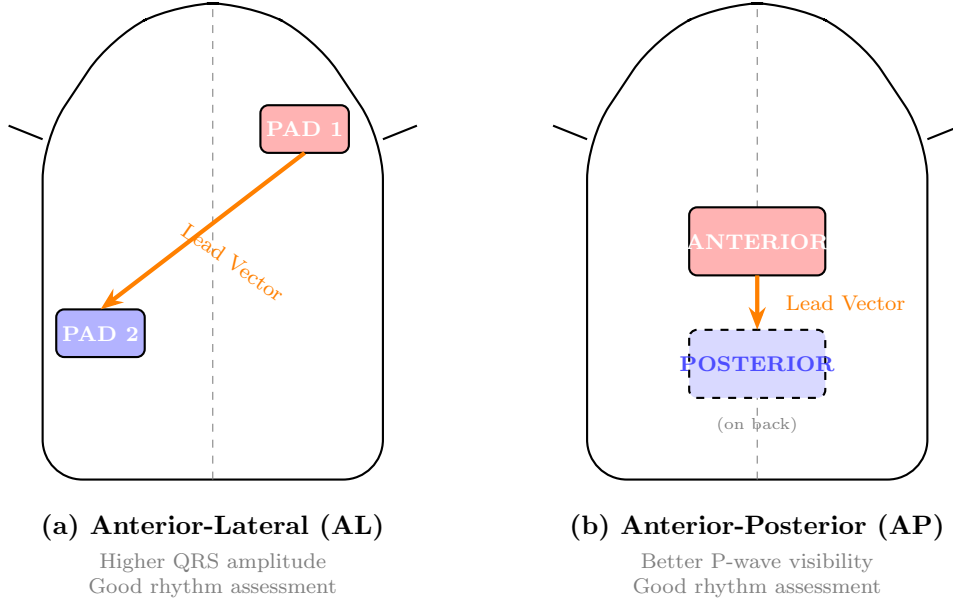


Figure 11: Defibrillation electrode placement configurations and corresponding lead vectors. (a) Anterior-Lateral: one pad below the right clavicle, one at the left mid-axillary line. (b) Anterior-Posterior: one pad on the precordium, one on the left infrascapular back. Neither configuration provides standard diagnostic lead morphology.

Table 6: ECG Morphological Considerations by Electrode Placement

Feature	Anterior-Lateral	Anterior-Posterior
P-wave visibility	Moderate	Good
QRS amplitude	High	Moderate
ST-segment reliability	Limited	Limited
Rhythm assessment suitability	Good	Good
Diagnostic morphology	Not reliable	Not reliable

It is clinically important to recognize that ECG signals acquired through defibrillation pads are suitable for *rhythm monitoring* (identifying shockable vs. non-shockable rhythms) but are generally *not appropriate for diagnostic morphological analysis* such as ST-segment evaluation for ischemia or precise axis determination [2]. Post-shock transient QRS voltage reduction may also be observed due to chest wall tissue edema or myocardial stunning, which should not be misinterpreted as a pathological change.

6.2 Signal Recovery Performance

The multi-stage signal processing pipeline achieves cumulative noise reduction through each processing stage:

Table 7: Expected SNR Improvement by Processing Stage

Processing Stage	SNR Improvement	Latency
Hardware protection + baseline restoration	N/A (enables recovery)	< 100 ms
Analog band-pass filtering	10 dB–15 dB	< 1 ms
50/60 Hz notch filtering	15 dB–25 dB	< 1 ms
Adaptive filtering (NLMS)	10 dB–15 dB	5 ms–20 ms
Wavelet denoising (db4, level 8)	5 dB–10 dB	10 ms–50 ms
Cumulative improvement	40 dB–65 dB	< 100 ms

The cumulative SNR improvement across all processing stages is visualized in Figure 12.

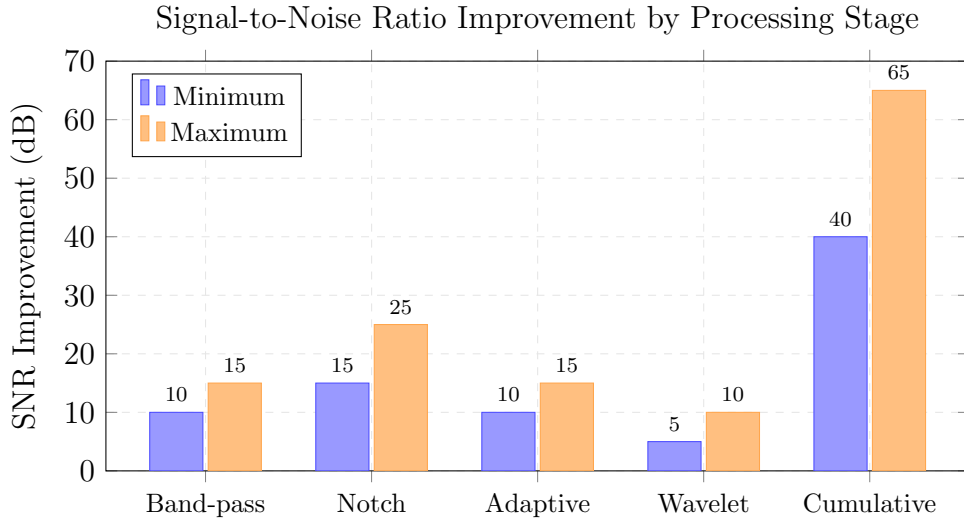


Figure 12: Signal-to-noise ratio (SNR) improvement contributed by each processing stage in the signal acquisition pipeline. The cumulative improvement of 40 dB–65 dB enables extraction of clinically useful ECG signals from heavily corrupted defibrillation recordings.

The total processing latency of less than 100 ms is well within clinical acceptability for real-time rhythm monitoring, where decision delays of up to 500 ms are generally tolerable [1].

6.3 Comparison with Existing Approaches

Current commercial defibrillator-monitors (e.g., Zoll X Series, Physio-Control LIFEPAK 15, Philips HeartStart MRx) implement basic through-pad ECG monitoring with hardware protection and simple digital filtering. The approach presented in this report extends beyond current commercial practice by:

1. Proposing **adaptive filtering** with reference-signal-based artifact cancellation rather than simple blanking or fixed-coefficient filtering.
2. Incorporating **wavelet denoising** as a complementary processing stage for residual artifact removal.

3. Evaluating **deep learning approaches** (CNN-LSTM) for automated rhythm classification directly from noisy pad-acquired signals.
4. Exploring **multifunctional sensor integration** to reduce the total sensor burden on critically ill patients.

7 Conclusion and Future Work

This report has presented a comprehensive analysis of the engineering challenges and solutions for measuring ECG signals through defibrillation electrodes. The key findings are:

1. **Electrode polarization** following defibrillation is the primary obstacle to rapid ECG recovery. Ag/AgCl electrodes conforming to ANSI/AAMI EC12 provide recovery to ≤ 400 mV within 4s, but active baseline restoration circuits are essential to minimize the “blind period.”
2. **Hardware protection** incorporating current-limiting resistors, TVS diodes, and switchable filter time constants enables the ECG monitoring circuit to survive defibrillation pulses exceeding 5 kV while complying with IEC 60601-2-4 safety requirements.
3. **Multi-stage digital signal processing** combining adaptive filtering (NLMS or RLS) with wavelet denoising can achieve cumulative SNR improvements of 40 dB–65 dB, enabling extraction of clinically useful ECG signals from heavily corrupted recordings.
4. **Defibrillation pad ECG** is suitable for rhythm monitoring but not for diagnostic morphological analysis, a distinction that clinicians must understand to avoid misinterpretation.

Future research directions include:

- **Real-time CNN-LSTM deployment:** Optimization of hybrid deep learning models for embedded deployment in defibrillator hardware, enabling sub-500 ms automated rhythm classification from pad-acquired signals.
- **Advanced electrode materials:** Investigation of novel electrode coatings (e.g., iridium oxide, conductive polymers) that may provide faster polarization recovery than conventional Ag/AgCl.
- **Closed-loop adaptive systems:** Development of self-tuning signal processing pipelines that automatically adjust filter parameters based on detected noise characteristics and signal quality metrics.
- **Clinical validation:** Prospective clinical trials comparing diagnostic accuracy of pad-acquired ECG versus standard 12-lead ECG for rhythm assessment during cardiac arrest.
- **Multifunctional electrode validation:** Engineering prototype development and biocompatibility testing of integrated multi-sensor defibrillation pads.

By addressing the limitations of defibrillation electrode placement through advanced signal processing and innovative electrode design, the dual functionality of these electrodes for both shock delivery and ECG monitoring can be fully realized, contributing to more efficient and accurate clinical practices in cardiac emergency care.

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